



Faculty of Science and Arts
Department of Chemistry
Biochemistry Lab (CHEM 266)

Experiment 8

ENZYMATIC COLORIMETRIC METHOD (END POINT)

APPLICATION ON BEERS LAW

Pre lab questions

1. What dose end point determination mean?
2. What's the difference between plasma and serum?
3. 1ml equals what in micro liters (μl)?
4. Which bloodborn diseases could be transferred accidentally through direct contact with contaminated blood?

Precautions

- You are going to measure concentration of certain substances in blood, you should be careful when handling blood as any other bodily fluid
- Do not handle blood or any other equipment without wearing gloves, and wearing your lab coats
- if you have any visible cuts cover them with clean wound dressers.
- Do not touch your eyes, mouth, face with your gloves
- Discard serum contaminated objects, in the specified yellow bags
- After removing your gloves wash your hands for 2 minutes with water a soap then sanitize them with hand sanitizer (Higeen) or 70% ethanol

Objectives

- 1- Determination of Triglycerides, Glucose, Cholesterol, and Albumin in a blood sample.
- 2- Apply Beers law to calculate concentration of these substances
- 3- Learn how to use Micropipettes

Introduction

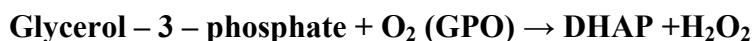
Many of the biochemical tests run routinely for patients and for checkups at hospitals are performed on body fluid, mostly on serum or plasma. Serum is the yellow watery part of blood that is left after blood has been allowed to clot and all blood cells have been removed. This is most easily done by. Plasma is in essence the same as serum, but is obtained by centrifuging the blood without clotting. Plasma is obtained by centrifugation before clotting occurs this is usually done by the use of an anticoagulant. The type of test required dictates what type of sample is used.

In this experiments we will apply two methods the enzymatic method to determine the concentration of triglyceride, glucose and cholesterol in serum and the colorimetric method to determine albumin concentration.

Part A: Triglycerides

Triglycerides are esters derived from glycerol and three fatty acids, they are the main constituents of the body fats in humans and animals as well as vegetable fats, and they are also present in the blood to enable bidirectional transference of adipose fat and blood glucose from the liver.

High levels of triglycerides in the blood stream have been linked to risk of heart diseases and heart stroke. So blood analysis for triglycerides should be performed in specific ages.



LPL = Lipoprotein Lipase

GK = Glycerol Kinase

GPO = Glycerol phosphate oxidase

DHAP = Dihydroxy phosphat oxidase

4 - **AA** = 4 - Amino antipyrine

POD = Peroxidase

FAA = Free Fatty Acids

λ_{max} = 500 nm

C_{Standard} = 200 mg / dl

Part B: Glucose

Glucose, a monosaccharide which we get from food we eat and our bodies use it for energy.

The healthy range is 90 – 130 mg/dl and after one or two hours it should be less than 180 mg/dl for the body to perform its function in a proper way.



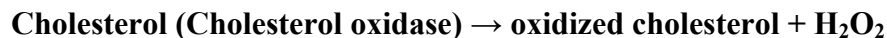
GOD = Glucose Oxidase

$$\lambda_{\text{max}} = 500 \text{ nm}$$

$$\text{C}_{\text{Standard}} = 100 \text{ mg / dl}$$

Part C: Cholesterol

Cholesterol is present in blood serum as cholesterol esters and free cholesterol. The cholesterol ester is hydrolyzed first by cholesterol oxidase to free cholesterol which is measured by oxidation with oxygen to form hydrogen peroxide which in turn reacts with 4- phenol and 4-amino antipyrine to form the red quinoneimine complex which its color intensity is proportional to the level of cholesterol present in the sample.



$$\lambda_{\text{max}} = 500 \text{ nm}$$

$$\text{C}_{\text{Standard}} = 200 \text{ mg / dl}$$

Part D: Albumin

Serum albumin binds with Bromocresol green indicator in acidic medium to form a green BCG complex, the amount of which is directly proportional to the albumin concentration present in the sample.

Albumin maintains water balance in serum and plasma and to transport a variety of ligands as calcium albumin and hormones.

This albumin test is non enzymatic.

$$\lambda_{\text{max}} = 630 \text{ nm}$$

$$\text{C}_{\text{Standard}} = 4.5 \text{ g / dl}$$

Procedure

A blood sample is collected in a plain tube, then it was centrifuged at 6000 RPM for 10 minutes the blood samples will be separated into different layers based on the density of its components

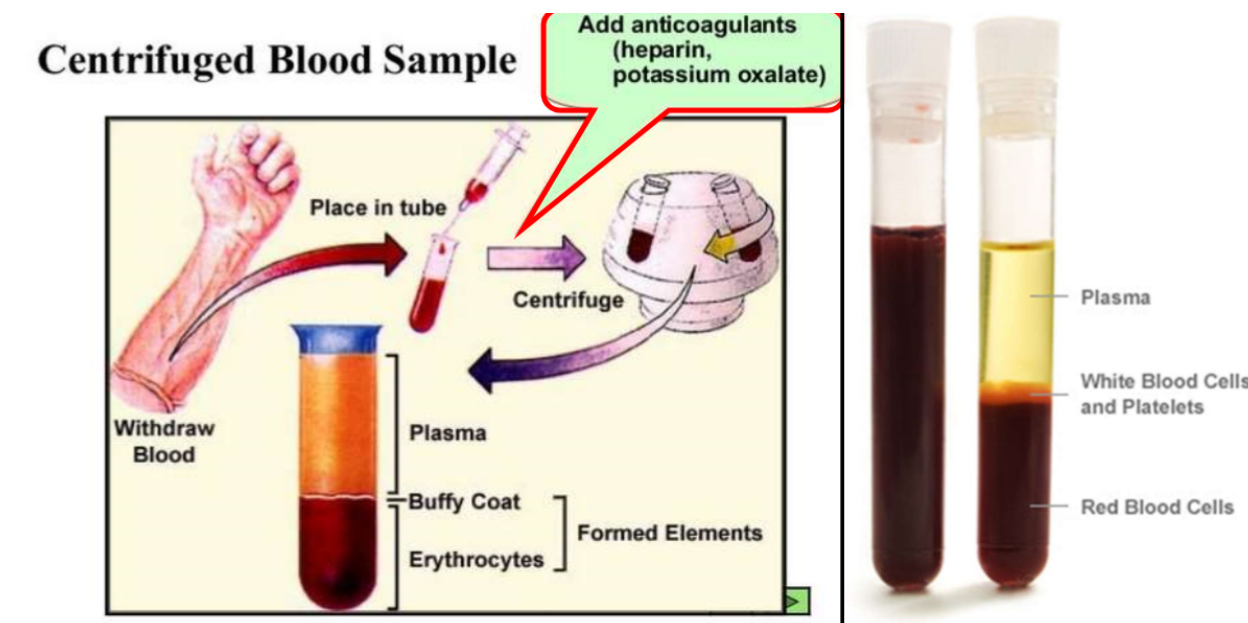


Figure (1) centrifugation of blood sample: (A) Shows the steps of blood collection. Type of anticoagulants used . (B) shows blood components separated into layers based on their densities the denser blood cells and platelets to the bottom of the centrifuge tube, leaving the liquid serum fraction resting above the packed cells

<http://d1yboe6750e2cu.cloudfront.net/i/53fa4d4076a212c2f80b9c4429cfc079753087a6>

1- Arrange test tubes according to the following table:

Test tube	Reagent	Serum Sample	Standard Solution
1	1.0 mL	—	—
2	1.0 mL	10 μ L	—
3	1.0 mL	—	10 μ L

2- Repeat the above table for each enzymatic and no enzymatic reaction.

3- Read absorbance for each at the proper or particular wave length.

- 4- Apply Beer's Law to determine the concentration of each of the required compounds in blood serum.

$$C_{\text{Sample}} = (A_{\text{Sample}} / A_{\text{Standard}}) \times C_{\text{Standard}}$$

Micropipettes will be utilized in to transfer the small quantity of liquid as (10 μ L) micropipettes differ in size, volume dispensed, and specific pipette tips required to aspirate liquid.

You should note that

*** note that the tip is the only part of the pipette that makes contact with the solution.

*** A new tip is utilized for every sample in order to prevent cross contamination.

A



Figure (2) (A) micropipette

B



(B) schematic view of the micropipette

References

https://www.biotek.com/assets/tech_resources/Monitoring_Glucose_Production_App_Note.pdf

- <http://d1yboe6750e2cu.cloudfront.net/i/53fa4d4076a212c2f80b9c4429cfc079753087a6>

- <https://www.biovision.com/documentation/datasheets/K554.pdf>



Faculty of Science and Arts
Applied Chemistry
Biochemistry Lab (CHEM 266)
Experiment 4
Enzymatic Colorimetric Method (End Point)

Report Author:

Lab Partner:

Date Due:

Date Turned In:

Unknown #:

Unknown Identification:

Purpose:

Data & Calculations:

Apply Beer's Law to calculate

A: Concentration of cholesterol =

B: Concentration of Glucose =

C: Concentration of Triglyceride =

D: Concentration of Albumin =

Conclusion:

Errors encountered running the experiment